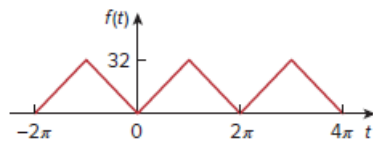


Problem

Find the Fourier series expansion of the function in Fig. 17.16.

Figure 17.16:



Step-by-step solution

Step 1 of 6

Refer to Figure 17.16 in the textbook.

The function $f(t)$ is expressed as,

$$f(t) = \begin{cases} \frac{32}{\pi}t, & 0 < t < \pi \\ -\frac{32}{\pi}t + 64, & \pi < t < 2\pi \end{cases}$$

Time period of $f(t)$ $T = 2\pi$.

Frequency of $f(t)$ is,

$$\begin{aligned} \omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{2\pi} \\ &= 1 \end{aligned}$$

Step 2 of 6

Calculation of a_0 .

$$\begin{aligned}
 a_0 &= \frac{1}{T} \int_0^T f(t) dt \\
 &= \frac{1}{2\pi} \int_0^{2\pi} f(t) dt \\
 &= \frac{1}{2\pi} \left[\int_0^{\pi} f(t) dt + \int_{\pi}^{2\pi} f(t) dt \right] \\
 &= \frac{1}{2\pi} \left[\int_0^{\pi} \left(\frac{32}{\pi} t \right) dt + \int_{\pi}^{2\pi} \left(-\frac{32}{\pi} t + 64 \right) dt \right] \\
 &= \frac{32}{2\pi} \left[\frac{1}{\pi} \int_0^{\pi} t dt - \frac{1}{\pi} \int_{\pi}^{2\pi} t dt + 2 \int_{\pi}^{2\pi} dt \right] \\
 &= \frac{32}{2\pi} \left[\frac{1}{\pi} \frac{t^2}{2} \Big|_0^{\pi} - \frac{1}{\pi} \frac{t^2}{2} \Big|_{\pi}^{2\pi} + 2t \Big|_{\pi}^{2\pi} \right] \\
 &= \frac{32}{2\pi} \left[\frac{1}{\pi} \left(\frac{\pi^2}{2} - 0 \right) - \frac{1}{\pi} \left(\frac{4\pi^2}{2} - \frac{\pi^2}{2} \right) + 2(2\pi - \pi) \right] \\
 &= 16 \left[\frac{1}{2} - \frac{3}{2} + 2 \right] \\
 &= 16
 \end{aligned}$$

Therefore, the value of a_0 is 4.

Step 3 of 6

Calculation of a_n .

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_0^T f(t) \cos n\omega_0 t dt \\
 &= \frac{2}{2\pi} \int_0^{2\pi} f(t) \cos nt dt \\
 &= \frac{1}{\pi} \left[\int_0^{\pi} f(t) \cos nt dt + \int_{\pi}^{2\pi} f(t) \cos nt dt \right] \\
 &= \frac{1}{\pi} \left[\int_0^{\pi} \left(\frac{32}{\pi} t \right) \cos nt dt + \int_{\pi}^{2\pi} \left(-\frac{32}{\pi} t + 64 \right) \cos nt dt \right] \\
 &= \frac{32}{\pi} \left[\frac{1}{\pi} \int_0^{\pi} t \cos nt dt - \frac{1}{\pi} \int_{\pi}^{2\pi} t \cos nt dt + 2 \int_{\pi}^{2\pi} \cos nt dt \right] \\
 &= \frac{32}{\pi} \left[\frac{1}{\pi} \left(t \frac{\sin nt}{n} \Big|_0^{\pi} - \frac{1}{n} \int_0^{\pi} \sin nt \right) - \frac{1}{\pi} \left(t \frac{\sin nt}{n} \Big|_{\pi}^{2\pi} - \frac{1}{n} \int_{\pi}^{2\pi} \sin nt \right) + 2 \frac{\sin nt}{n} \Big|_{\pi}^{2\pi} \right] \\
 &= \frac{32}{\pi} \left[\frac{1}{\pi} \left(\pi \frac{\sin n\pi}{n} - 0 \frac{\sin n0}{n} + \frac{1}{n^2} \cos nt \Big|_0^{\pi} \right) \right. \\
 &\quad \left. - \frac{1}{\pi} \left(2\pi \frac{\sin n2\pi}{n} - \pi \frac{\sin n\pi}{n} + \frac{1}{n^2} \cos nt \Big|_{\pi}^{2\pi} \right) + 2 \frac{\sin nt}{n} \Big|_{\pi}^{2\pi} \right]
 \end{aligned}$$

Step 4 of 6

Continue the calculations.

$$\begin{aligned}
 &= \frac{32}{\pi} \left[\frac{1}{\pi} \left(0 - 0 + \frac{1}{n^2} (\cos n\pi - \cos n0) \right) \right. \\
 &\quad \left. - \frac{1}{\pi} \left(0 - 0 + \frac{1}{n^2} \cos n2\pi - \cos n\pi \right) + \frac{2}{n} \sin n2\pi - \sin n\pi \right] \\
 &= \frac{32}{\pi} \left[\frac{1}{\pi n^2} ((-1)^n - 1) - \frac{1}{\pi n^2} (1 - (-1)^n) + \frac{2}{n} (0 - 0) \right] \\
 &= \frac{64}{\pi^2 n^2} ((-1)^n - 1)
 \end{aligned}$$

Therefore, the value of a_n is $\frac{64}{\pi^2 n^2} ((-1)^n - 1)$.

Step 5 of 6

Calculation of b_n .

$$\begin{aligned}
 b_n &= \frac{2}{T} \int_0^T f(t) \sin n\omega_0 t \, dt \\
 &= \frac{2}{2\pi} \int_0^{2\pi} f(t) \sin nt \, dt \\
 &= \frac{1}{\pi} \left[\int_0^{\pi} f(t) \sin nt \, dt + \int_{\pi}^{2\pi} f(t) \sin nt \, dt \right] \\
 &= \frac{1}{\pi} \left[\int_0^{\pi} \left(\frac{32}{\pi} t \right) \sin nt \, dt + \int_{\pi}^{2\pi} \left(-\frac{32}{\pi} t + 64 \right) \sin nt \, dt \right] \\
 &= \frac{32}{\pi} \left[\frac{1}{\pi} \int_0^{\pi} t \sin nt \, dt - \frac{1}{\pi} \int_{\pi}^{2\pi} t \sin nt \, dt + 2 \int_{\pi}^{2\pi} \sin nt \, dt \right] \\
 &= \frac{32}{\pi} \left[\frac{1}{\pi} \left(-t \frac{\cos nt}{n} \Big|_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos nt \, dt \right) - \frac{1}{\pi} \left(-t \frac{\cos nt}{n} \Big|_{\pi}^{2\pi} + \frac{1}{n} \int_{\pi}^{2\pi} \cos nt \, dt \right) - 2 \frac{\cos nt}{n} \Big|_{\pi}^{2\pi} \right] \\
 &= \frac{32}{\pi} \left[\frac{1}{\pi} \left(-\pi \frac{\cos n\pi}{n} + 0 \frac{\cos n0}{n} + \frac{1}{n^2} \sin nt \Big|_0^{\pi} \right) \right. \\
 &\quad \left. - \frac{1}{\pi} \left(-2\pi \frac{\cos n2\pi}{n} + \pi \frac{\cos n\pi}{n} + \frac{1}{n^2} \sin nt \Big|_{\pi}^{2\pi} \right) - \frac{2}{n} (\cos n2\pi - \cos n\pi) \right] \\
 &= \frac{32}{\pi} \left[\frac{1}{\pi} \left(-\frac{\pi}{n} (-1)^n + 0 + \frac{1}{n^2} (\sin n\pi - \sin n0) \right) \right. \\
 &\quad \left. - \frac{1}{\pi} \left(-\frac{2\pi}{n} + \frac{\pi}{n} (-1)^n + \frac{1}{n^2} (\sin n2\pi - \sin n\pi) \right) - \frac{2}{n} (1 - (-1)^n) \right] \\
 &= \frac{32}{\pi} \left[\frac{1}{\pi} \left(-\frac{\pi}{n} (-1)^n \right) - \frac{1}{\pi} \left(-\frac{2\pi}{n} + \frac{\pi}{n} (-1)^n \right) - \frac{2}{n} (1 - (-1)^n) \right] \\
 &= \frac{32}{\pi} \left[-\frac{1}{n} (-1)^n + \frac{2}{n} - \frac{1}{n} (-1)^n - \frac{2}{n} + \frac{2}{n} (-1)^n \right] \\
 &= \frac{32}{\pi} \left[-\frac{2}{n} (-1)^n + \frac{2}{n} (-1)^n \right] \\
 &= 0
 \end{aligned}$$

Therefore the value of b_n is 0.

Step 6 of 6

The function $f(t)$ can be expressed as

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Substitute 16 for a_0 , $\frac{64}{\pi^2 n^2}((-1)^n - 1)$ for a_n and 0 for b_n in the above equation.

$$\begin{aligned} f(t) &= 16 + \sum_{n=1}^{\infty} \left[\left(\frac{64}{n^2 \pi^2} (-1)^n - \frac{64}{n^2 \pi^2} \right) \cos nt + 0 \sin nt \right] \\ &= 16 + \sum_{n=1}^{\infty} \left[\left(\frac{64}{n^2 \pi^2} (-1)^n - \frac{64}{n^2 \pi^2} \right) \cos nt \right] \\ &= \left[16 + \left[\frac{64}{\pi^2} (-1)^1 - \frac{64}{\pi^2} \right] \cos t + \left[\frac{64}{2^2 \pi^2} (-1)^2 - \frac{64}{2^2 \pi^2} \right] \cos 2t \right. \\ &\quad \left. + \left[\frac{64}{3^2 \pi^2} (-1)^3 - \frac{64}{3^2 \pi^2} \right] \cos 3t + \left[\frac{64}{4^2 \pi^2} (-1)^4 - \frac{64}{4^2 \pi^2} \right] \cos 4t + \dots \right] \\ &= \left[16 - \frac{128}{\pi^2} \cos t + (0) \cos 2t - \frac{128}{3^2 \pi^2} \cos 3t + (0) \cos 4t - \frac{128}{5^2 \pi^2} \cos 5t \right. \\ &\quad \left. + (0) \cos 6t + \dots \right] \\ &= 16 - \frac{128}{\pi^2} \cos t + 0 + -\frac{128}{3^2 \pi^2} \cos 3t + 0 - \frac{128}{5^2 \pi^2} \cos 5t + 0 + \dots \\ &= 16 - \frac{128}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{n^2} \cos nt, \quad n = 2k-1 \end{aligned}$$

Therefore, the Fourier series of function $f(t)$ is $\boxed{16 - \frac{128}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{n^2} \cos nt, \quad n = 2k-1}.$